

CHESTER, CT, USA

WMO: 720545

Lat: 41.384N

Lon: 72.506W

Elev: 417

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00169

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.9	12.3	-9.1	3.4	12.5	-5.6	4.1	17.7	22.8	28.6	20.4	27.9	8.3	330	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	88.4	74.1	85.9	72.9	82.4	71.0	77.5	84.1	75.9	81.7	74.5	79.5	6.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	134.5	79.9	73.5	126.4	77.6	72.9	123.8	77.0	41.2	83.0	39.6	82.4	38.4	79.6	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.9	16.7	14.6	DB	1.2	93.2	5.6	3.8	-2.9	96.0	-6.2	98.2	-9.4	100.4	-13.5	103.2
(5)				WB	-0.4	79.4	5.3	2.0	-4.3	80.9	-7.4	82.0	-10.4	83.1	-14.3	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	28.9	32.3	39.0	49.3	59.7	67.5	74.2	72.5	64.9	54.6	44.0	36.2
	DBStd	17.05	9.46	9.29	8.44	7.56	7.39	6.08	4.99	4.88	6.66	7.95	8.06	9.04
	HDD50	2307	654	499	354	99	7	0	0	0	0	43	215	436
	HDD65	5505	1118	916	805	473	200	42	3	3	85	335	631	894
	CDD50	3045	0	3	14	79	307	525	749	696	447	185	33	7
	CDD65	772	0	0	0	4	35	118	287	234	82	12	0	0
	CDH74	5192	0	0	0	37	290	792	2120	1512	408	33	0	0
	CDH80	1409	0	0	0	7	76	192	666	379	85	4	0	0
Wind	WSAvg	6.4	7.3	7.5	7.9	7.2	6.1	5.6	5.1	4.8	5.2	6.1	6.8	6.7
Precipitation	PrecAvg	47.8	3.5	3.1	4.2	4.0	3.7	4.1	3.7	4.3	4.4	4.3	3.9	4.4
	PrecMax	59.7	6.9	8.4	9.4	8.3	7.3	10.2	9.3	9.8	10.2	15.3	6.6	7.9
	PrecMin	35.0	1.9	1.5	0.6	1.3	0.8	0.5	1.9	1.9	1.1	0.7	1.2	1.4
	PrecStd	6.8	1.4	1.3	2.0	1.8	1.5	2.6	1.5	2.2	2.4	3.1	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	60.8	69.6	80.6	87.9	90.1	93.0	90.5	87.9	78.6	66.3	61.2
		MCWB	55.1	53.5	59.4	63.4	70.8	73.1	77.4	77.1	75.8	66.0	59.0	59.1
	2%	DB	52.1	54.2	62.5	71.7	81.4	84.5	89.7	86.4	81.8	72.9	63.2	56.7
		MCWB	49.8	50.3	52.1	57.3	67.0	70.4	75.0	74.4	70.7	67.0	57.9	54.8
	5%	DB	46.2	49.9	55.3	65.6	76.7	81.5	86.3	83.9	79.0	70.2	60.6	54.4
		MCWB	43.5	46.0	47.8	54.7	63.6	68.6	73.2	72.4	69.3	65.2	55.4	52.8
	10%	DB	43.0	45.4	52.0	62.6	72.4	79.0	83.6	81.6	75.1	66.2	56.7	50.5
		MCWB	39.1	40.5	45.0	51.3	60.9	66.9	72.4	71.2	68.6	62.1	52.0	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	56.8	60.4	63.9	72.5	76.5	80.0	79.7	77.3	72.2	63.5	60.7
		MCDB	56.7	58.5	66.1	71.4	83.3	83.2	89.2	86.3	84.4	75.7	64.6	60.9
	2%	WB	49.8	51.0	54.1	60.8	69.7	74.0	77.5	77.5	74.4	69.1	60.8	55.3
		MCDB	50.4	52.9	57.7	68.0	77.2	80.8	84.7	83.6	78.5	70.9	62.2	55.7
	5%	WB	44.8	46.0	49.2	57.7	67.2	72.1	76.0	75.9	72.9	66.2	56.4	53.5
		MCDB	46.0	48.8	56.2	62.8	73.5	78.4	82.6	80.7	76.4	69.1	58.4	54.3
	10%	WB	39.1	41.0	45.8	54.5	64.5	70.1	74.7	74.2	71.6	63.3	53.1	47.8
		MCDB	42.8	45.1	49.7	59.2	69.0	75.3	80.3	78.6	74.6	65.0	56.0	50.0
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.9	14.1	17.0	17.5	16.8	16.0	15.7	16.5	15.7	14.1	12.3
		MCDBR	16.8	17.9	18.7	21.8	23.5	19.3	18.4	17.4	17.7	16.7	16.7	15.7
		MCWBR	17.0	14.6	11.7	11.9	11.8	9.5	7.7	7.8	9.3	10.2	13.6	15.8
	5% WB	MCDBR	17.3	18.4	17.4	17.6	20.2	16.9	16.1	14.7	14.3	14.5	15.8	16.0
		MCWBR	18.9	16.8	13.4	12.3	11.7	9.3	7.7	7.4	9.1	10.9	15.4	17.7
Clear-Sky Solar Irradiance	taub		0.300	0.309	0.331	0.368	0.414	0.455	0.475	0.445	0.377	0.338	0.310	0.298
	taud		2.494	2.475	2.464	2.386	2.282	2.204	2.159	2.234	2.402	2.524	2.558	2.555
	Ebn at Noon		272	287	292	286	274	262	255	259	272	273	267	265
	Edh at Noon		23	28	31	36	42	45	47	42	33	26	22	21
All-Sky Solar Radiation	RadAvg		594	866	1177	1531	1711	1859	1893	1687	1391	933	651	493
	RadStd		49	78	92	165	170	187	104	95	108	88	51	45

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+103

Nomenclature: See separate page